



Data Center Best Practices with DGX B300

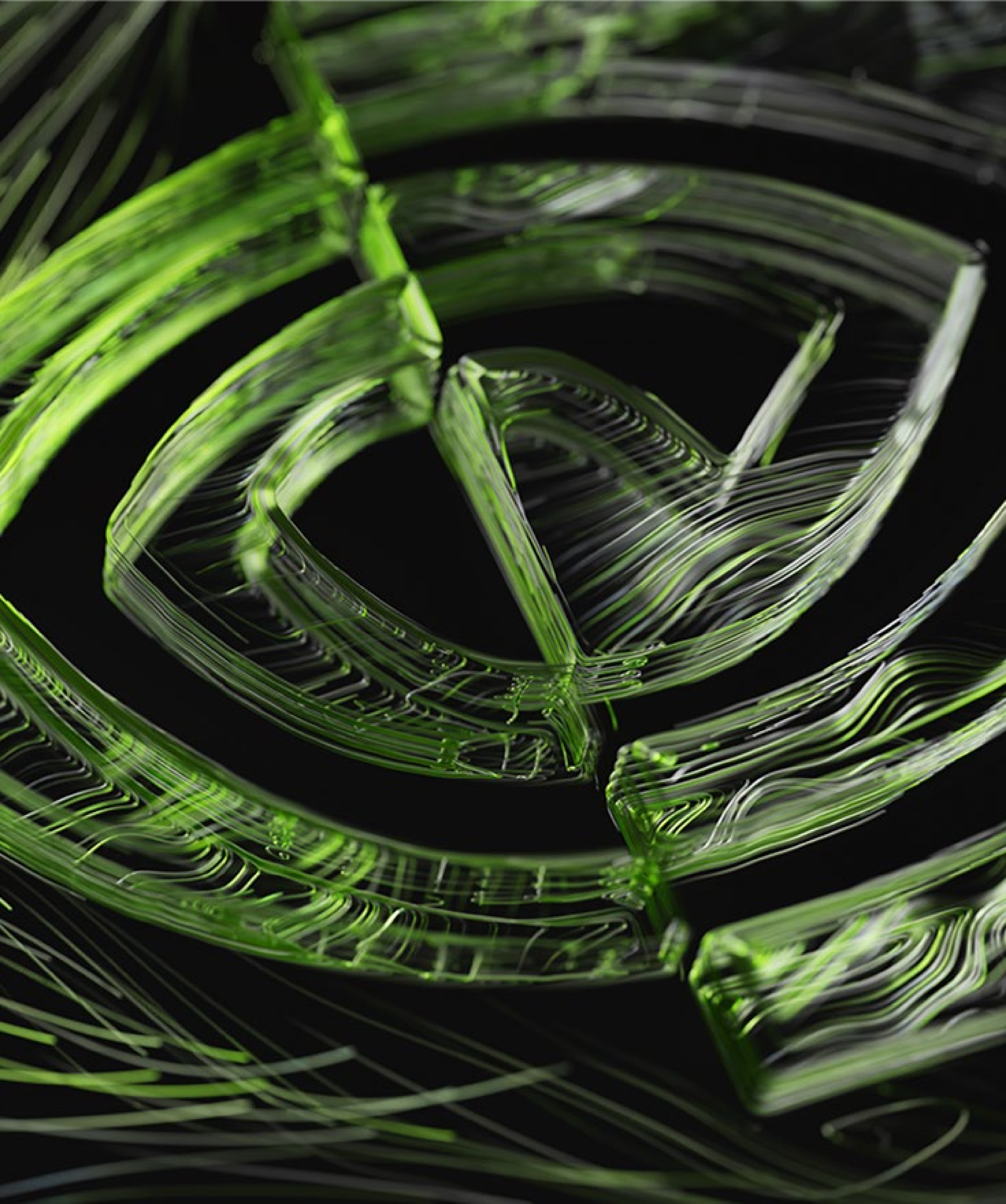
Steven Hambruch DCA, Distinguished Engineer, Data Center Infrastructure Architect

Dennis O'Brien DCEP, CISSP, RCDD, Senior Data Center Deployment Engineer

Version 1.0



Revision History			
Version	Date	Author	Description
1.0	02-27-2026	Steven Hambruch Dennis O'Brien	Initial Release



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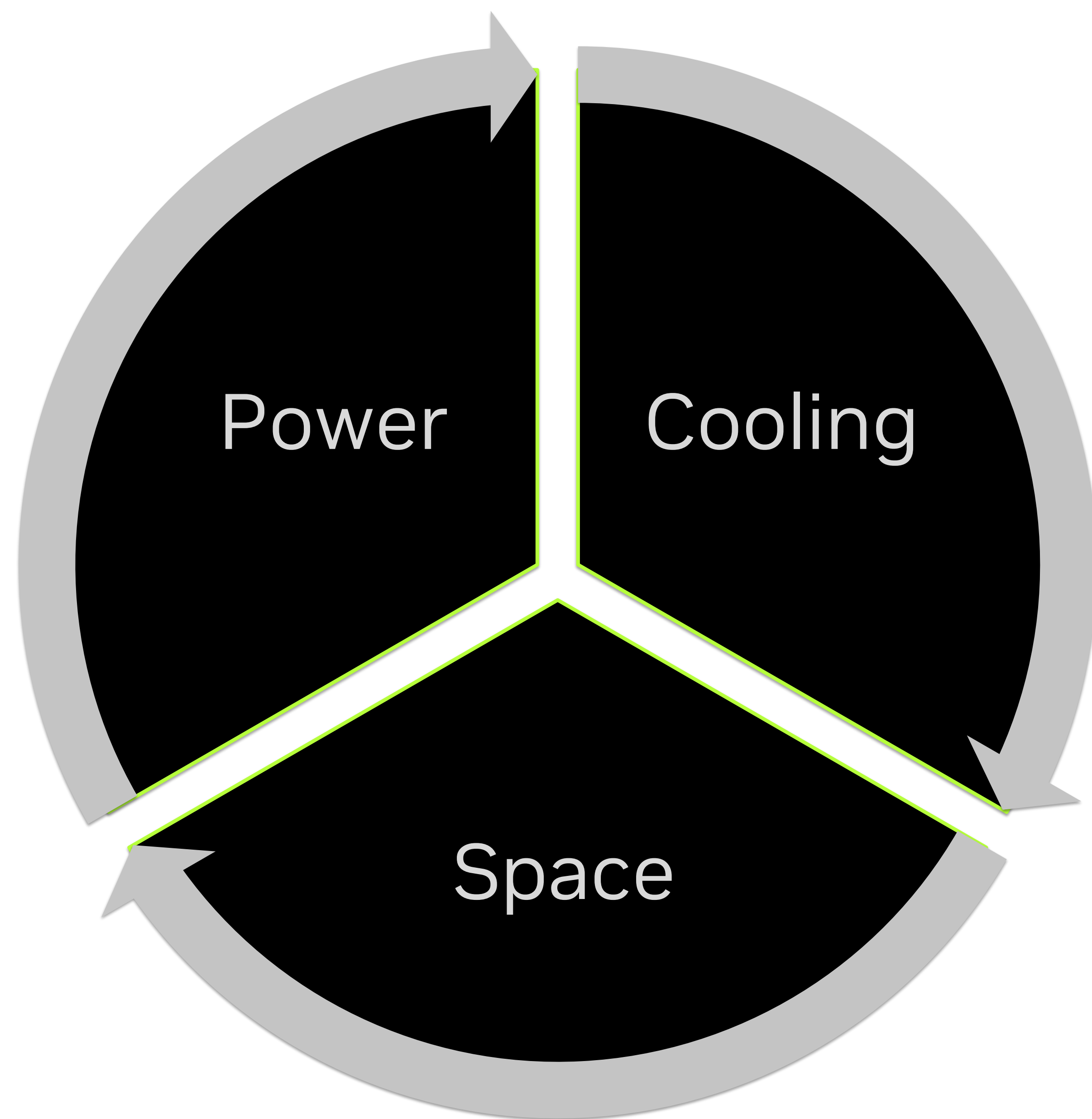
The Economy of Data Center Resources

Deploying high density workloads in the Data Center



The Economy of Data Center Resources

Deploying High Density Workloads in the Data Center



- Data Centers have finite resources
 - These limited resources create scarcity in the face of demand
 - This drives the need to optimize resource utilization for efficiency
- The three main resource constraints are
 - Power
 - Cooling
 - Space
- These resources are interrelated
 - Excess demand on one resource could negatively impact other resources
 - These interrelationships can also impact InfiniBand cable lengths



Introduction to DGX B300 Data Center Architecture

Overview of DGX Hardware, Connectivity,
and Resource Requirements

DGX B300

One universal building block for the AI data center



DGX B300 System Specifications		
	DC Busbar Version***	AC Rack PDU
Estimated System Power	14.5 kW**	15 kW**
Estimated Peak System Power	19 kW**	19.7 kW**
Estimated System Weight	>271.2 lbs. (123 kgs)*	>370.4 lbs. (168 kgs)*
System Dimensions	10 Rack Units (RU) Height: 17.5 in (444 mm) Width: 19.0 in (482.2 mm) Length: 35.3 in (897.1 mm) * Does not include Heat Exchanger	10 Rack Units (RU) Height: 17.5 in (444 mm) Width: 19.0 in (482.2 mm) Length: 35.3 in (897.1 mm)
Operating Temperature	10°C to 35°C (50°F to 95°F)	10°C to 35°C (50°F to 95°F)
Cooling Media	Active Rear Door Heat Exchanger	Air
Maximum Operating Altitude Above Sea Level	3048 meters (10,000 feet) maximum 10° to 30°C: altitude 0 – 1000 ft 10° to 25°C: altitude 0 – 5000 ft 10° to 20°C: altitude 0 – 10000 ft	3048 meters (10,000 feet) maximum 10° to 30°C: altitude 0 – 1000 ft 10° to 25°C: altitude 0 – 5000 ft 10° to 20°C: altitude 0 – 10000 ft
Estimated Acoustic Noise Operating	Acoustic Power (LWA,m)** 25 °C /77 °F = 95 dB	Acoustic Power (LWA,m)** 25 °C /77 °F = 92 dB

* Estimated. Actual specification may vary
** Estimated based on 25°C Supply Air Temperature
*** Does not include requirement for the power shelf (space, weight, power), requires DGX power shelf

DGX B300

Rack Specifications



High Density 4 B300 Systems Per Rack	
Power Source	DC Busbar with Power Shelves Four 1U 33kW DC Power Shelves per Rack
Power Demand*	58kW Average 76kW Peak
Rack Dimensions (Excluding Rear Door Heat Exchanger)	48U Height 2236mm (88 Inches) Width 600mm (24 Inches) Depth 1068mm (42 Inches)
Rack Weight* (Excluding Rear Door Heat Exchanger)	1 122kg (2474 lbs)
Cooling Media	Air Cooling with Active Rear Door Heat Exchanger
Supply Temperature	Inlet 10-35°C (50-95°F) Delta T at 25°C SAT: 20.6°C (69°F)
Cooling Demand*	6200cfm

*Estimated. Actual specification may vary



Low Density 2 B300 Systems Per Rack	
Power Source	AC Rack PDUs
Power Demand*	30kW Average 39.4kW Peak
Rack Dimensions (Excluding Rear Door Heat Exchanger)	48U Height 2236mm (88 Inches) Width 600mm (24 Inches) Depth 1068mm (42 Inches)
Rack Weight* (Excluding Rear Door Heat Exchanger)	1301 kg (1 170 lbs)
Cooling Media	Air Cooling
Supply Temperature	Inlet 10-35°C (50-95°F) Delta T 18°C (64°F)
Cooling Demand*	4200cfm

*Estimated. Actual specification may vary

DGX B300 SuperPOD Data Center Infrastructure

High Density Rack Architecture

- DGX B300 systems are available in both DC and AC versions
- For high density deployments (4 systems per rack), systems are deployed in an MGX style rack, which features an integrated rear DC Busbar and is powered by DC Power Shelves, instead of traditional AC Rack PDUs
- Active Rear Door Heat Exchangers (RDHx) are recommended to help manage heat loads in high density deployments
- Lower density deployments (2 systems per rack) may use traditional AC Rack PDUs, and unassisted air cooling.



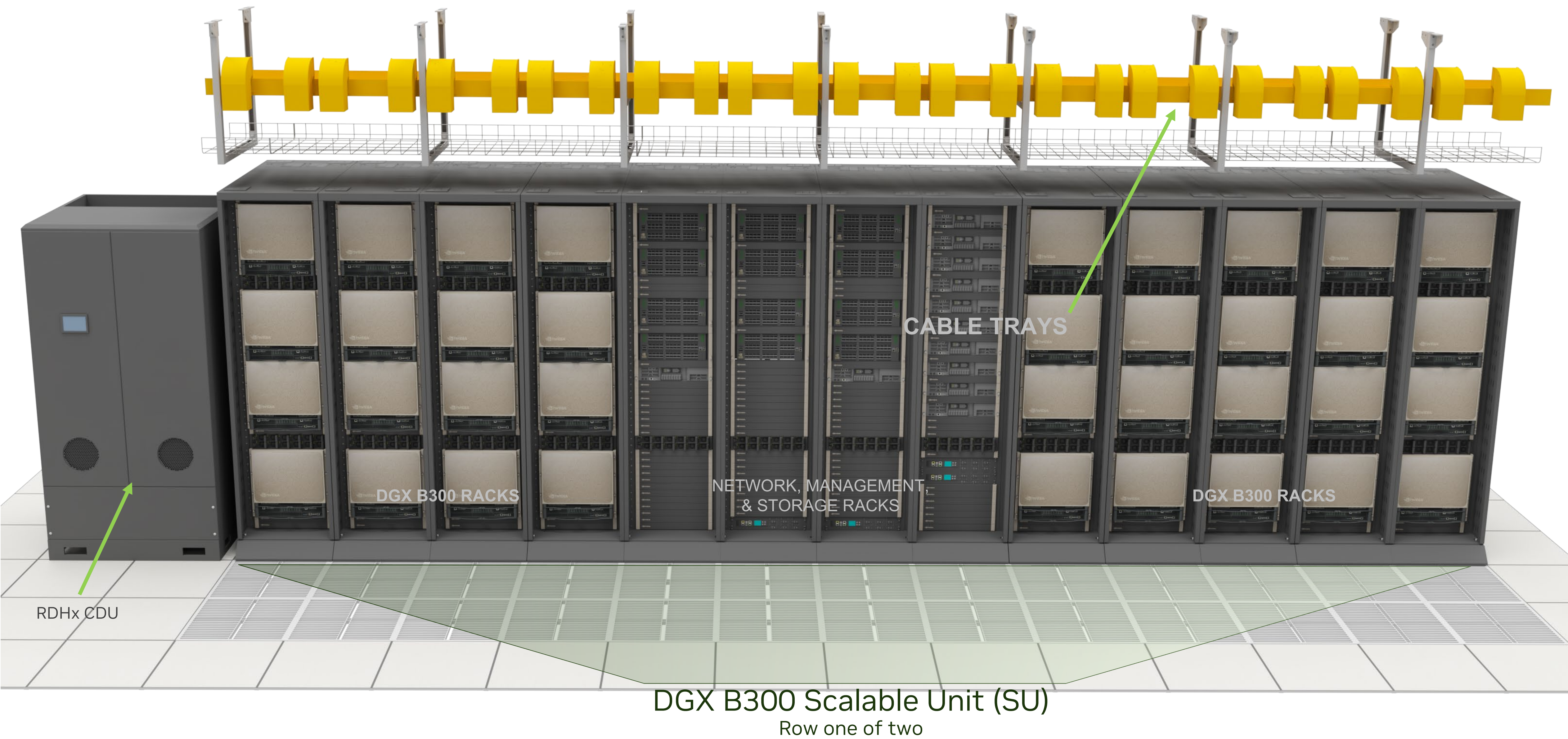
DGX B300 systems deployed in MGX style rack with integrated DC Busbar and power shelves, rear view of rack



Example of Active Rear Door Heat Exchangers

DGX B300 SuperPOD Data Center Infrastructure

Data Center Deployment Pattern



- A DGX Scalable Unit (SU) consists of up to 72 (XDR) or 64 (Spectrum-X) DGX B300 systems
- A DGX SuperPOD can be comprised of multiple Scalable Units, interconnected with XDR IB or Spectrum-X as well as Ethernet In-band and Storage Fabrics, combined with a compatible storage system.
- Performance and cost optimized
- High Density and Low-Density Deployment options
 - Four air-cooled DGX B300s per 48U MGX rack in high density deployment pattern, with Active Rear Door Heat Exchangers, or
 - Two traditional air-cooled DGX B300s per 48U/52U rack
- Proximity of the cabinets is governed by cable distance limits
 - Compute Fabric (XDR IB or Spectrum-X) drives rack-to-rack cable distance requirements

Power Demand for a typical 64 node high density SU

Component	Qty	Total Average Power	Total Peak Power
DGX B300 Racks	16	820Kva	1228Kva
Network Management & Storage Racks	8	104Kva	105Kva
CDUs	2	24Kva	70Kva
RDHx Units	24	27Kva	47Kva
Totals		975Kva	1450Kva

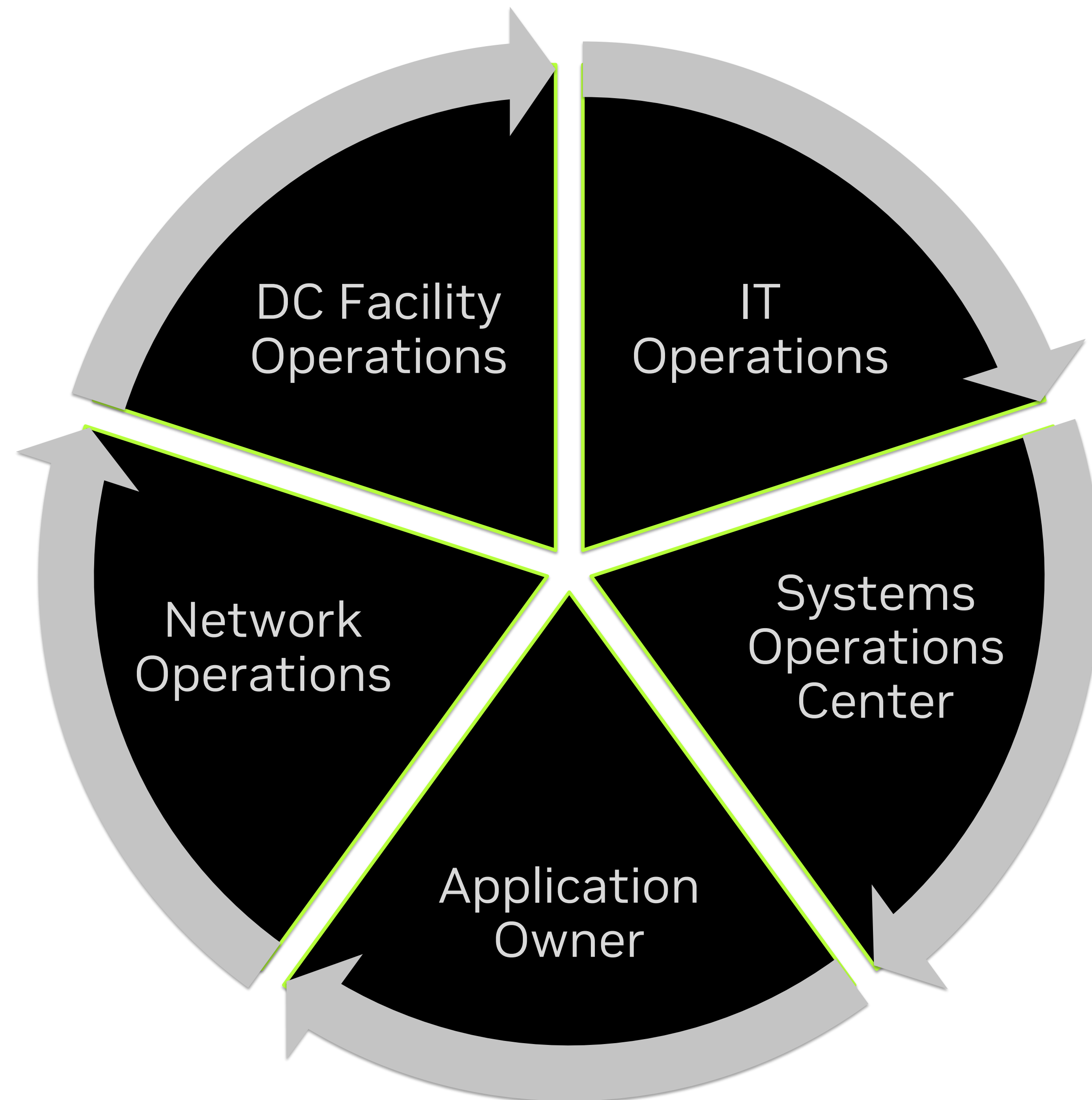
Planning a Data Center Deployment

Overview of recommended planning and
logistics steps



Planning a data center deployment

Equal consideration must be given to multiple domains in a DGX B300 deployment



- A well-planned deployment will involve aligning multiple constituencies, including
 - Data Center Facility Operations
 - IT Operations
 - Systems Operations Center
 - Application Owner
 - Network Operations
- Solutions to resource constraints in one domain may impact planning, operations, personnel, and budgets in another domain

Planning a data center deployment

Site Assessments and Space Planning

- To prepare for a DGX B300 Deployment, the following should be assessed and integrated into the deployment plan
 - Pre-deployment Site Survey and Space Planning
 - Does the site comply with all the operating requirements of the equipment?
 - Assign and pre-prepare the location where the systems will be deployed
 - Loading Dock Assessment
 - Does the site have a loading dock with hydraulic levelers for smooth transition from truck to staging area?
 - Staging Space Assessment
 - Is there sufficient secure staging space to unpack or uncrate the equipment and prepare it for deployment?
 - Pathway Assessment
 - Ramps, Door Dimensions, Elevators, Turn Radiuses, Etc.
 - Assure there are no obstacles that would impede the safe and smooth movement of equipment to the deployment area



Planning a data center deployment

Infrastructure Planning and Deployment

- Preparing Deployment location for a B300 Deployment
 - Power Provisioning
 - TCS and FWS Cooling Loop Provisioning
 - CDU Installation
 - Aisle Containment Structures
 - Cable Management Structures
 - Seismic Bracing (If required)



Planning a data center deployment

Safety

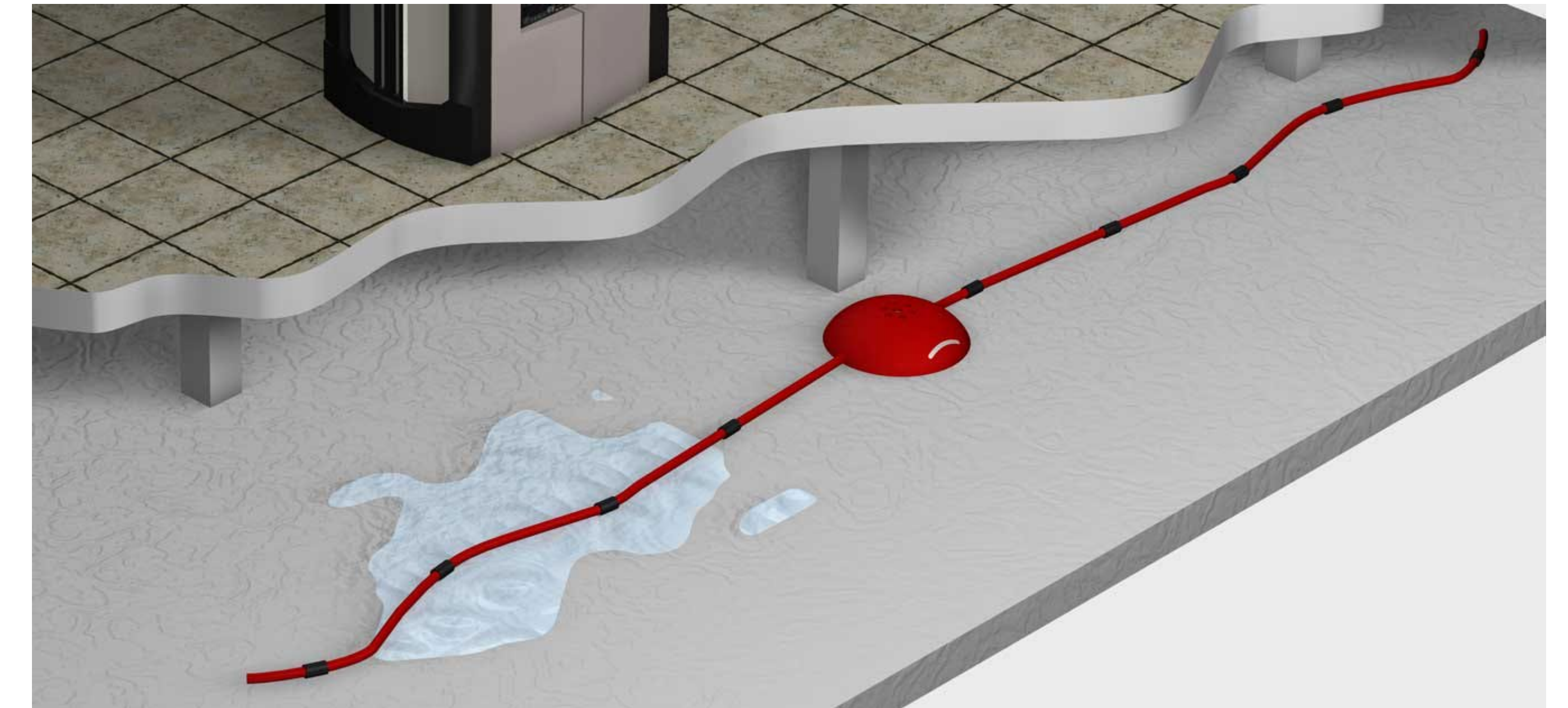
- Assure worker compliance with all safety protocols and protective measures
 - Use of rack tugs, server lifts, pallet jacks, or other conveyances in accordance with manufacturer recommendations
 - Use of Personal Protective Equipment (PPE)
 - Safety shoes, hard hats, safety glasses, gloves, lumbar support belts
 - Hearing protection
 - Proper and timely waste disposal
 - Personnel Safety Training Program compliance



Planning a data center deployment

Safety

- Where applicable, assure that the following are in place
 - Leak Detection Systems
 - Spill Mitigation Training
 - PG25 Safety Program
 - PG25 Compliant Personal Protective Equipment (PPE)
 - PG25 Spill Kits
 - PG25 Storage
 - Proper PG25 Disposal procedures
- Note: Consult with local or national Authorities Having Jurisdiction for safety and regulatory requirements when deploying PG25 based cooling systems.



Rear Door Heat Exchangers

Overview

- NVIDIA Does not provide or recommend specific RDHx models. Consult with suppliers to assure compliance with heat load and airflow requirements
 - Passive Rear Door Heat Exchangers are not recommended for DGX B300 Systems
- Rear Door Heat Exchangers (RDHx) units should be installed in accordance with manufacturer specifications, and in compliance with all Authorities Having Jurisdiction (AHJs)
- RDHx Design Considerations
 - Power Provisioning Requirements
 - Floor load calculations
 - Containment aisle pitch
 - Hot Aisle width and service clearances
 - Under floor vs. overhead piping designs
 - Supplemental fluid valve controls
 - Leak detection system
 - CDU placement
 - Grounding & Bonding
 - Local seismic requirements

Note: The Rear Door Heat Exchanger manufacturer may need to provide a custom mounting bracket for use with the MGX cabinet.

Rear Door Heat Exchangers

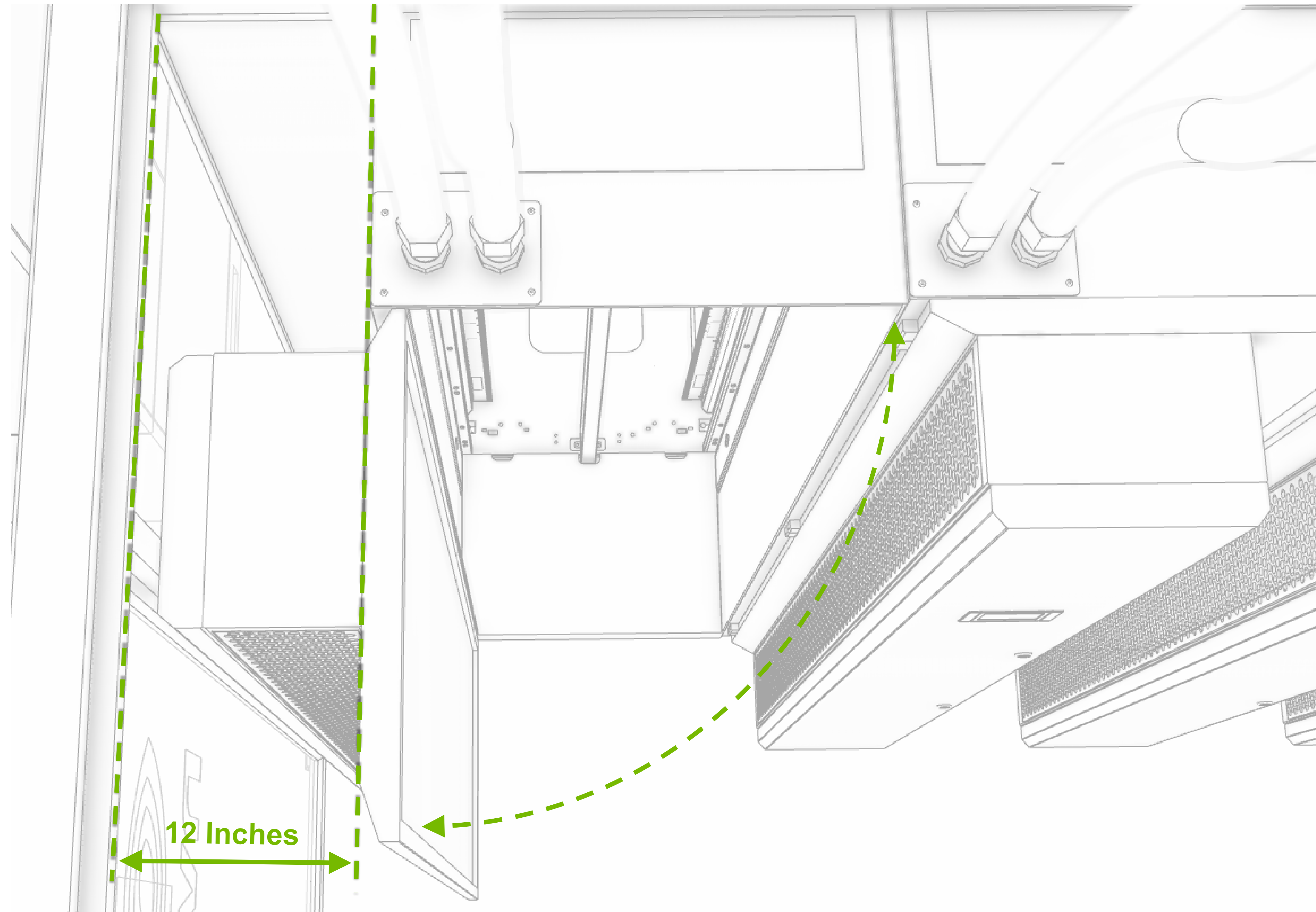
Overview

- RDHx Fluid Considerations
 - Fluid piping designs
 - Fluid connection types
 - Fluid system commissioning
 - Fluid management
 - Monitoring fluid levels & flow
 - Routine testing
 - Chemical analysis
- CDU Management
 - Routine inspections
 - Remote network monitoring & management
 - Data protocols, In-Band/Out of Band networks
 - CDU on-site training for operations Staff
 - Spare fluid and filters

Rear Door Heat Exchanger

Service Clearance

- For installations where Rear Door Heat Exchangers are used in conjunction with hot aisle containment, assure that the containment structure allows sufficient clearance for the heat exchanger door to open fully.

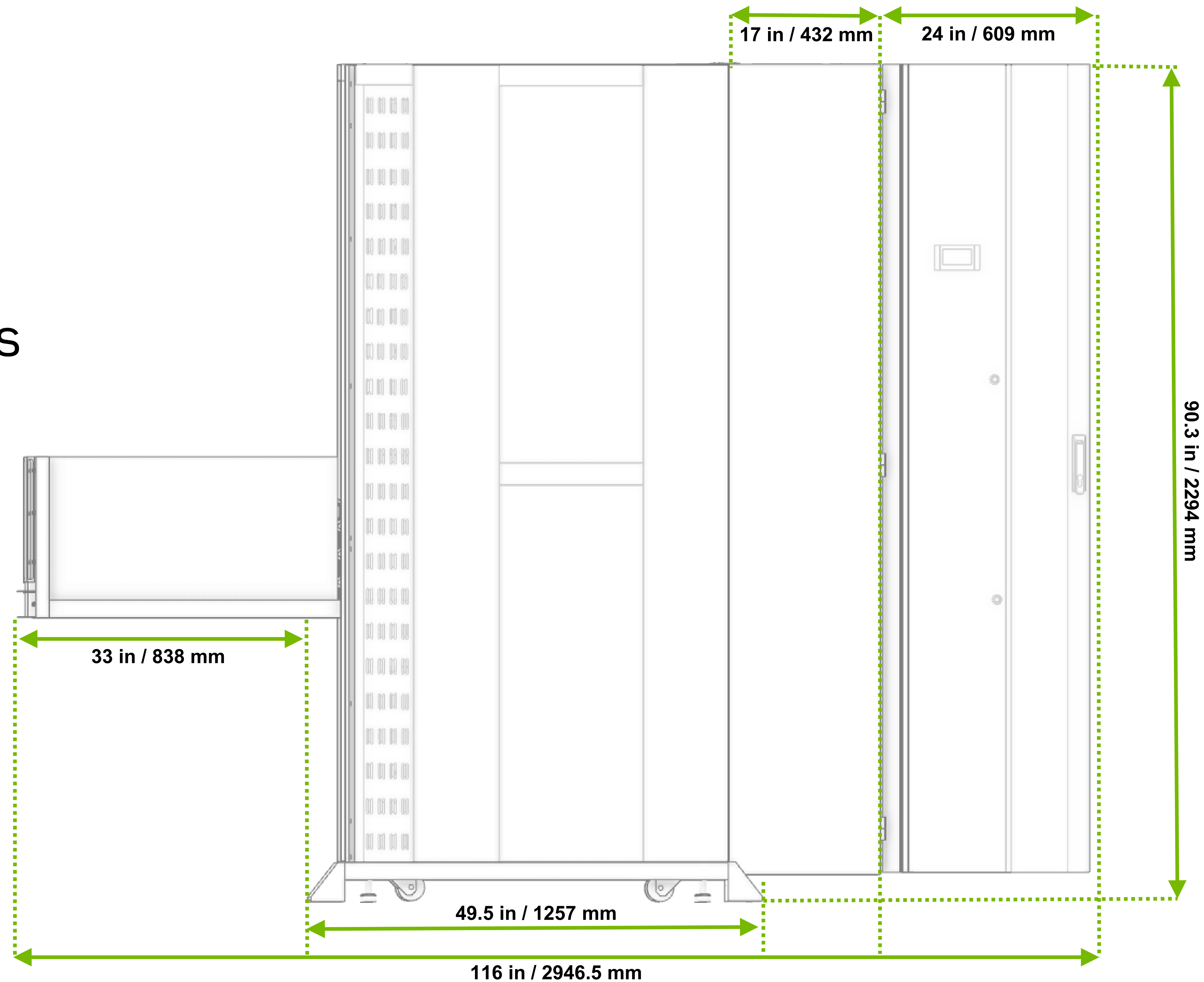


Rear Door Heat Exchanger Dimensions vary. Typical example depicted

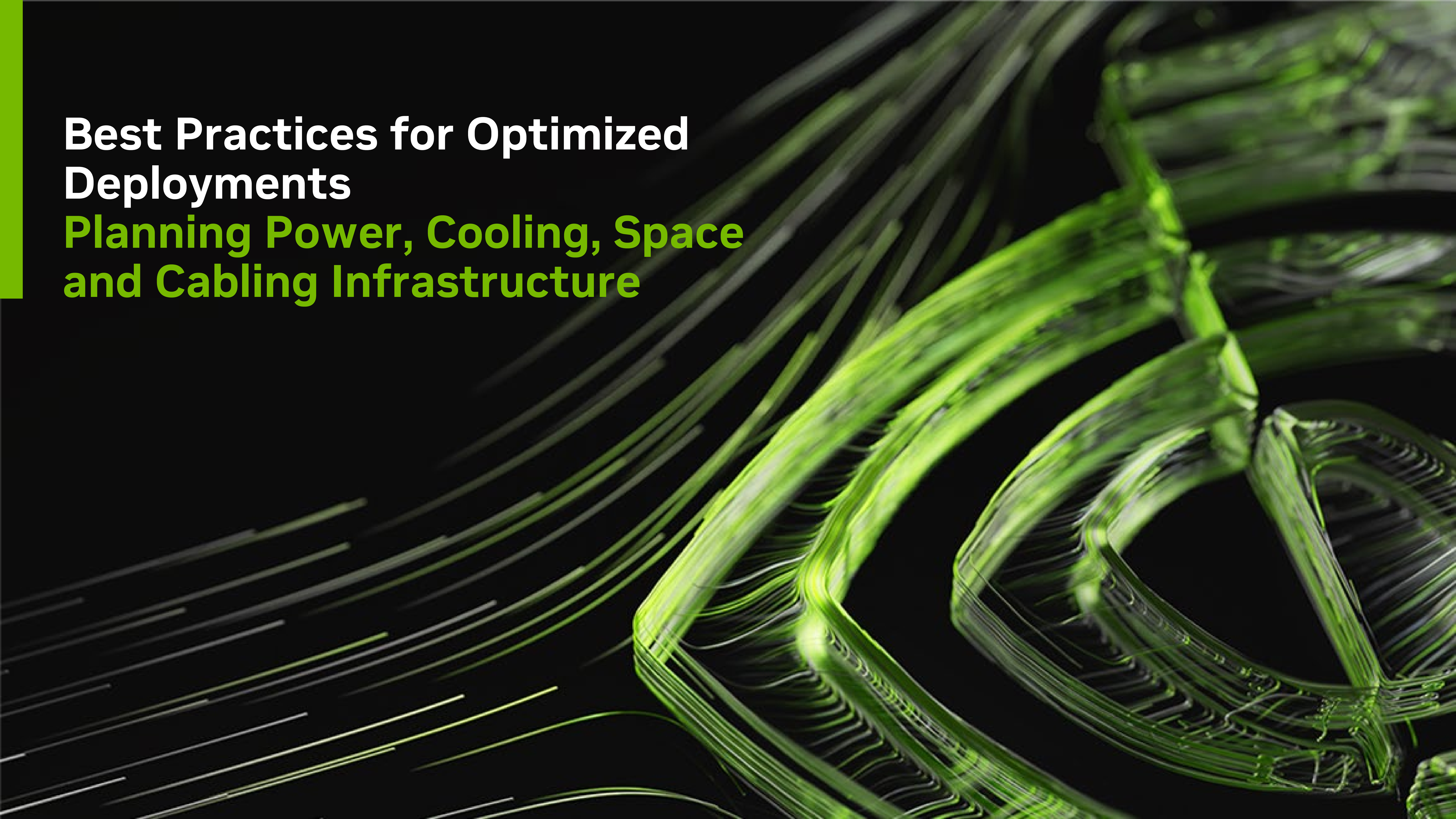
Rear Door Heat Exchanger

Service Clearance

- MGX Rack Service Clearances
 - Front of rack clearance to IT component servicing
 - Rear of rack clearance for servicing



Rear Door Heat Exchanger Dimensions vary. Typical example depicted

The background of the slide is a dark, abstract composition. It features numerous thin, glowing green lines that sweep across the frame, creating a sense of motion and energy. In the lower right quadrant, there are several bundles of fiber optic cables, their internal structures visible and glowing with a bright green light. The overall aesthetic is high-tech and futuristic.

Best Practices for Optimized Deployments

Planning Power, Cooling, Space and Cabling Infrastructure

DGX B300

Power Provisioning Specifications



High Density Deployment Pattern 4 DC B300 Systems Per Rack	
Power Source	DC Busbar with Power Shelves Four 1U 33kW DC Power Shelves per Rack
Power Demand*	58kW Average 76kW Peak
Power Shelf Redundancy	N+1 (4 B300 Systems requires at least 3 active Power Shelves)
Power Circuit Redundancy Per Power Shelf	2N (Each Shelf has redundant power inputs)
Circuit Provisioning Requirements	4 60amp Three Phase Circuits per utility source (400V minimum)



Low Density Deployment Pattern 2 AC B300 Systems Per Rack	
Power Source	AC Rack PDUs
Power Demand*	30kW Average 39.4kW Peak
Rack PDU Redundancy	2N (each B300 System has Redundant PSUs)
Circuit Provisioning Requirements	1 60amp Three Phase Circuit per utility source (400V Minimum)

Power Provisioning

Common distribution schemes compatible with High Density DGX B300 racks

Phase	Distribution Voltage	Line Voltage	Amps	Breaker Derating***	Number of Circuits Provisioned Per Utility Source	Rack Power Capacity kVA*	Maximum Supported DGX B300 Systems per Rack
3Φ	400	240	63	80.0%	4	141	4
3Φ	415	240	60	80.0%	4	139	4
3Φ	380	219	60	100.0%	4	160	4
3Φ	400	230	63	100.0%	4	176	4
3Φ	415	240	60	100.0%	4	174	4
* 0.99 power factor.							
*** Article 210.20 (A) of the NEC (NFPA 70) in the United States requires the rating of the overcurrent device to be 125% of the continuous load. Therefore, the maximum usable circuit capacity is 80% of the breaker's rated capacity. A similar requirement applies in Canada.							

Power Provisioning

Common distribution schemes compatible with Low Density DGX B300 racks

Phase	Distribution Voltage	Line Voltage	Amps	Breaker Derating	Number of Circuits Provisioned Per Utility Source	Rack Power Capacity kVA*	Maximum Supported DGX B300 Systems per Rack
3Φ	380	219	60	100.0%	1	39.9	2
3Φ	400	230	63	100.0%	1	44.1	2
3Φ	415	240	60	100.0%	1	43.6	2
* 0.99 power factor.							

Strategies for Cooling Optimization and Scalability

Environmental Parameters

- Each DGX B300 system requires up to approximately 2145 CFM of supply air at sea level
- Supply air should generally comply with ASHRAE Class A2 specifications, but with a maximum supply air temperature of 30°C (86°F).
- For optimal cooling efficiency and system performance, air contamination should be mitigated through filtration
- Minimum of MERV 13 (or EN779-2012 M6/F7, or ISO 16890 ePM1-50%) rated filter
- Rack area should meet the cleanliness level of the ISO 14644-1 Class-8 standard
 - Maximum particulate counts not to exceed 3,520,000 @ 0.5 µm/m3 for longer than 15 minutes.

ASHRAE specifications				
Range	Class	Dry-Bulb Temperature	Humidity Range, Non-Condensing	Maximum Dew Point
Recommended	All A	64.4–80.6 °F 18–27 °C	41.9 °F to 60% RH and 59 °F DP 5.5 °C to 60% RH and 15 °C DP	59 °F 15 °C
Allowable up to 30 °C for DGX B300 Systems	A1	59–89.6 °F 15–32 °C	20–80% RH	62.6 °F 17 °C
	A2	50–95 °F 10–35 °C	20–80% RH	69.8 °F 21 °C
	A3	41–104 °F 5–40 °C	10.4 °F DP and 8–85% RH -12 °C DP and 8–85% RH	75.2 °F 24 °C
	A4	41–113 °F 5–45 °C	10.4 °F DP and 8–90% RH -12 °C DP and 8–90% RH	75.2 °F 24 °C
	B	41–95 °F 5–35 °C	8–80% RH	82.4 °F 28 °C
Allowable per ASHRAE for various other classes of data center and telecom environments	C	41–104 °F 5–40 °C	8–80% RH	82.4 °F 28 °C

ISO 14644-1 standard for air cleanliness classifications						
Class	Particle Size ¹					
	> 0.1 µm	> 0.2 µm	> 0.3 µm	> 0.5 µm	> 1 µm	> 5 µm
1	10	2				
2	100	24	10	4		
3	1,000	237	102	35	8	
4	10,000	2,370	1,020	352	83	
5	100,000	23,700	10,200	3,520	832	29
6	1,000,000	237,000	102,000	35,200	8,320	293
7				352,000	83,200	2,930
8				3,520,000	832,000	29,300
9				35,200,000	8,320,000	293,000

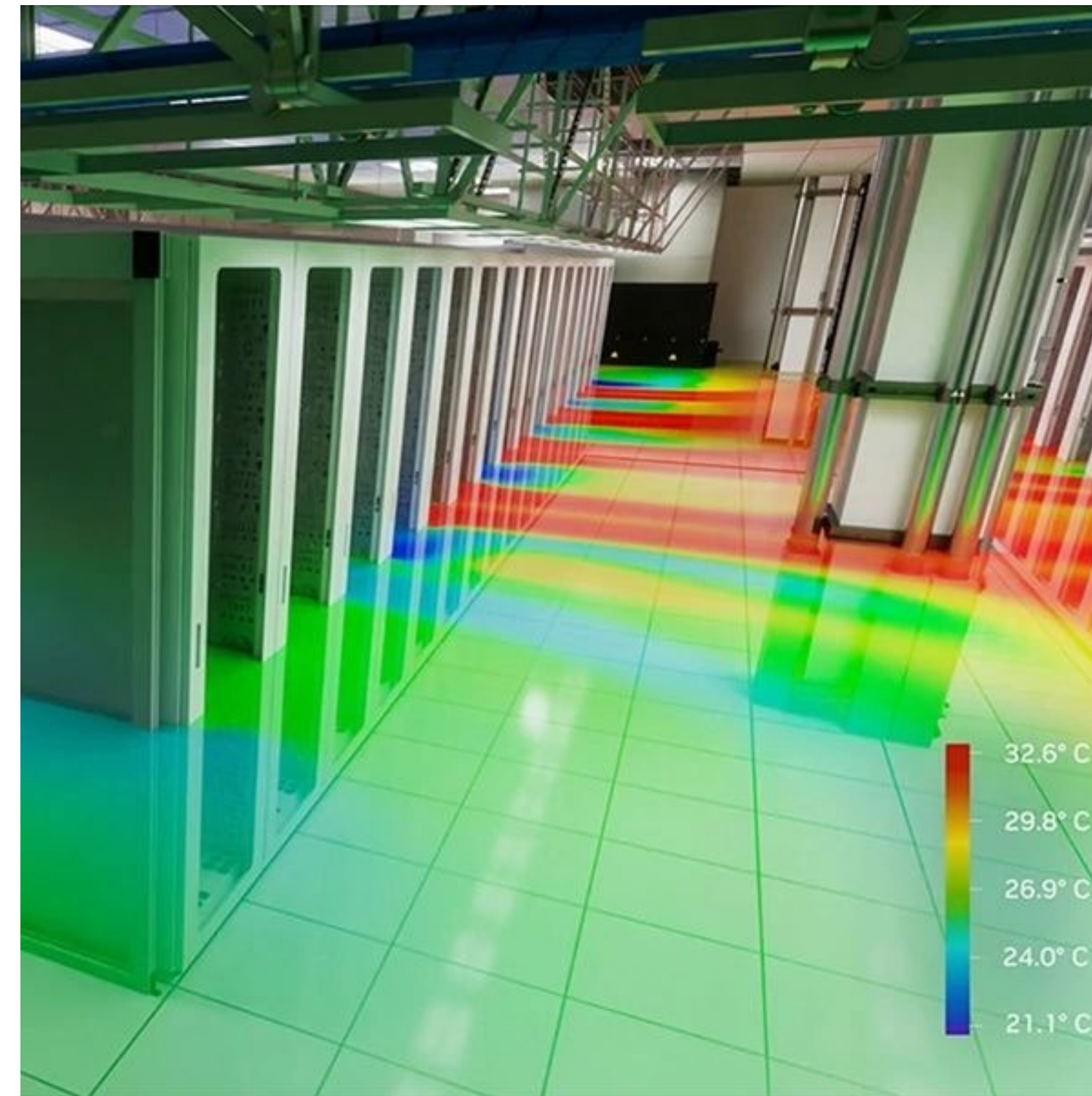
Uncertainties related to the measurement process require that data with no more than three significant figures be used in determining the classification level.

Strategies for Cooling Optimization and Scalability

For Plenum Raised Access Floor Environments



Aisle Containment Systems



Computational Fluid Dynamics



Blanking Panels



Brush Grommet

- Start with a solid foundation
 - Computational Fluid Dynamics (CFD) Modeling and Analysis
 - Aisle Containment
 - Blanking Panels
 - Brush Grommets
 - Humidification / Dehumidification
 - Proper air filtration
 - Perforated Floor Tile flow rate audits with anemometer

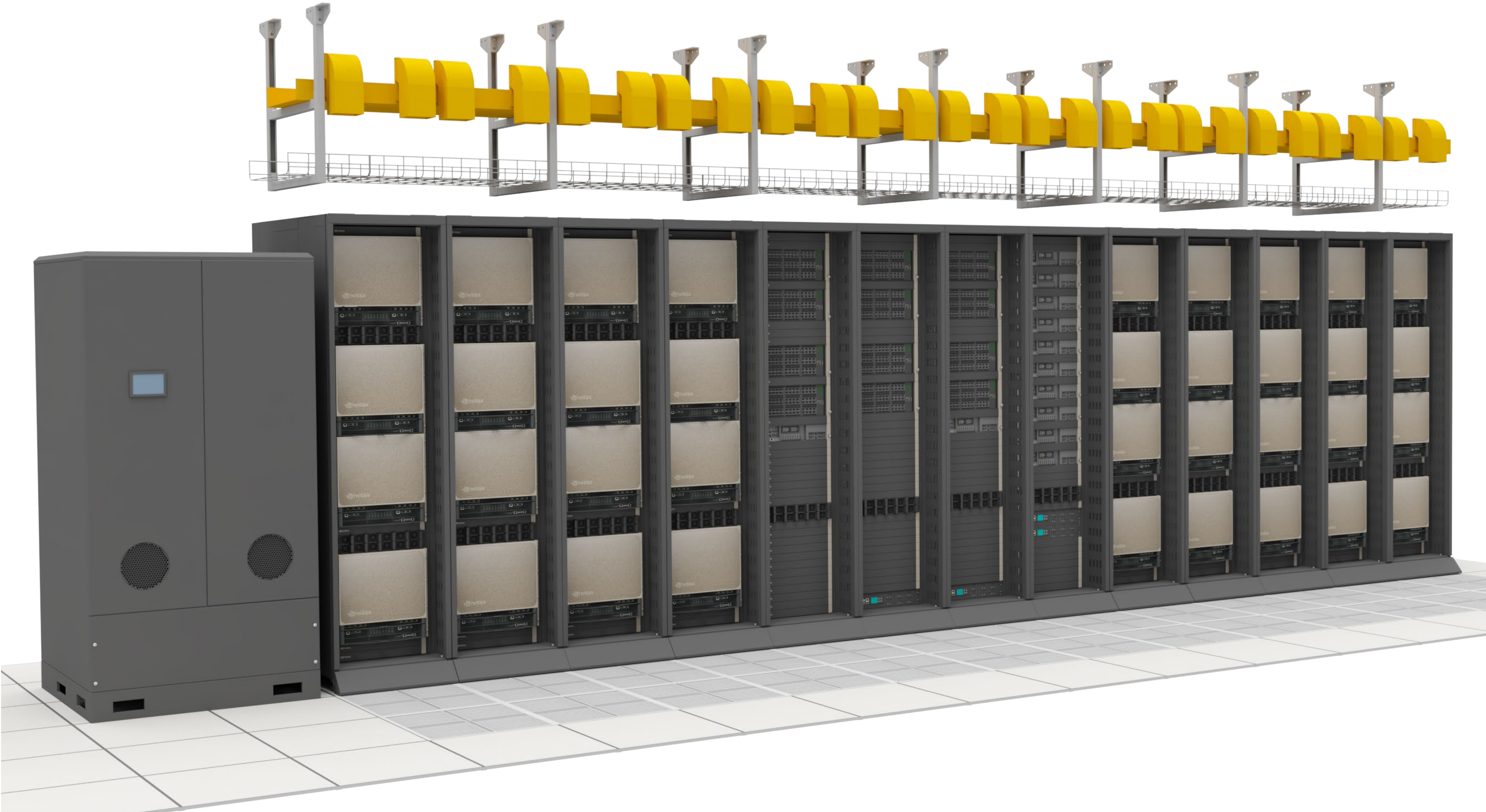


Perforated Floor Tile Flow Rate Analysis

Cabling

DGX Infrastructure Support

SU Count	Cluster Size # of nodes	Cluster Size # of GPUs	Leaf Switch Count	Spine Switch Count	Node-Leaf	Leaf-Spine
1	72	576	8	4	576	576
2	144	1152	16	8	1152	1152
4	288	2304	32	18	2304	2304
8	576	4608	64	36	4608	4608

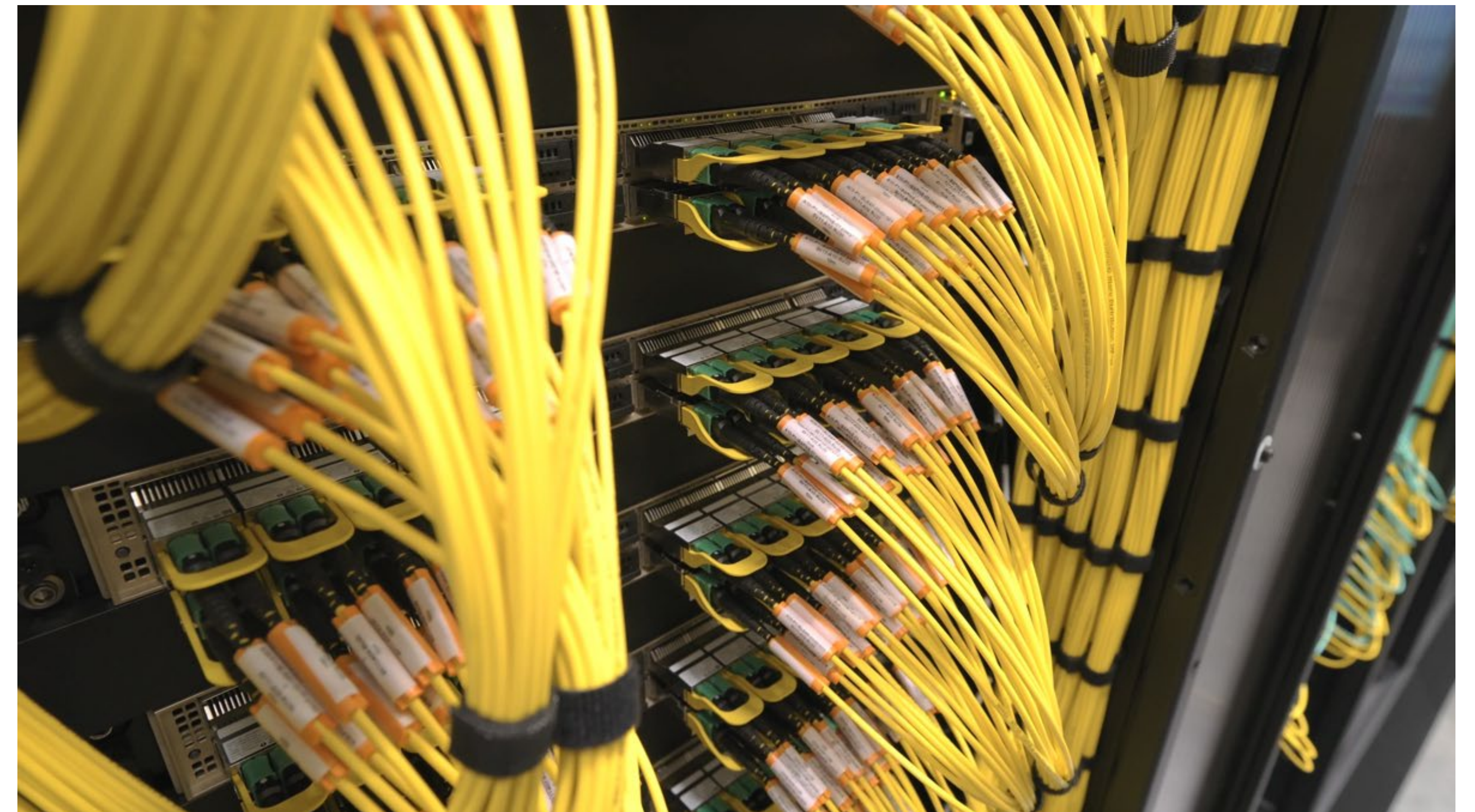


- There is a large concentration of cables in the central Management Racks and in the supporting cable pathways overhead. Pre-planning for the cabling infrastructure is a critical part of the design process.
- Areas of analysis include:
 - Top of rack cable penetration access
 - Cable tray types
 - Cable tray width and depth
 - Cable tray fill ratios
 - Structural load/weight planning
 - Cable management devices

Cabling

DGX Infrastructure Support

- Bundle cables together in groups of relevance (for example, ISL (Inter-Switch Link) cables and uplinks to core devices) to ease management and troubleshooting.
- Use cables of correct length.
- Allow for bend radiuses and intra-rack cable routing in cable length calculations
- Keep cable runs under 90% of the max distance supported for each media type, as specified in the applicable standard.
- Keep copper and fiber bundles and runs separated.
- Install spare cables in advance for future replacement of damaged cables.
- Use color coding of the cable ties. The colors should indicate the endpoints. Place labels at both ends, as well as along the run.



Example of proper cable management

Cabling

DGX Cabling Types

- The DGX B300 SuperPOD Compute Fabric is mainly interconnected with MPO12 APC passive single mode fiber cables connected to twin port OSFP Transceivers
- Copper Cat6 Ethernet cables are also used in the Management Network Fabric
- For maximum performance, cable lengths should be limited to 500 Meters or less.



Switch Side:
Twin-port 2x800GbE/XDR
OSFP IHS DR4



DGX Side:
Single Port 800GbE/XDR
OSFP RHS DR4

OSFP Shell Terms

IHS = Integrated **H**eat **S**ink – Cooling fins integrated into plug “*Finned Top, Twin-Port OSFP*”

RHS = **R**iding **H**eat **S**ink - on top of board connector cage “*Flat Top, Twin-Port OSFP*”

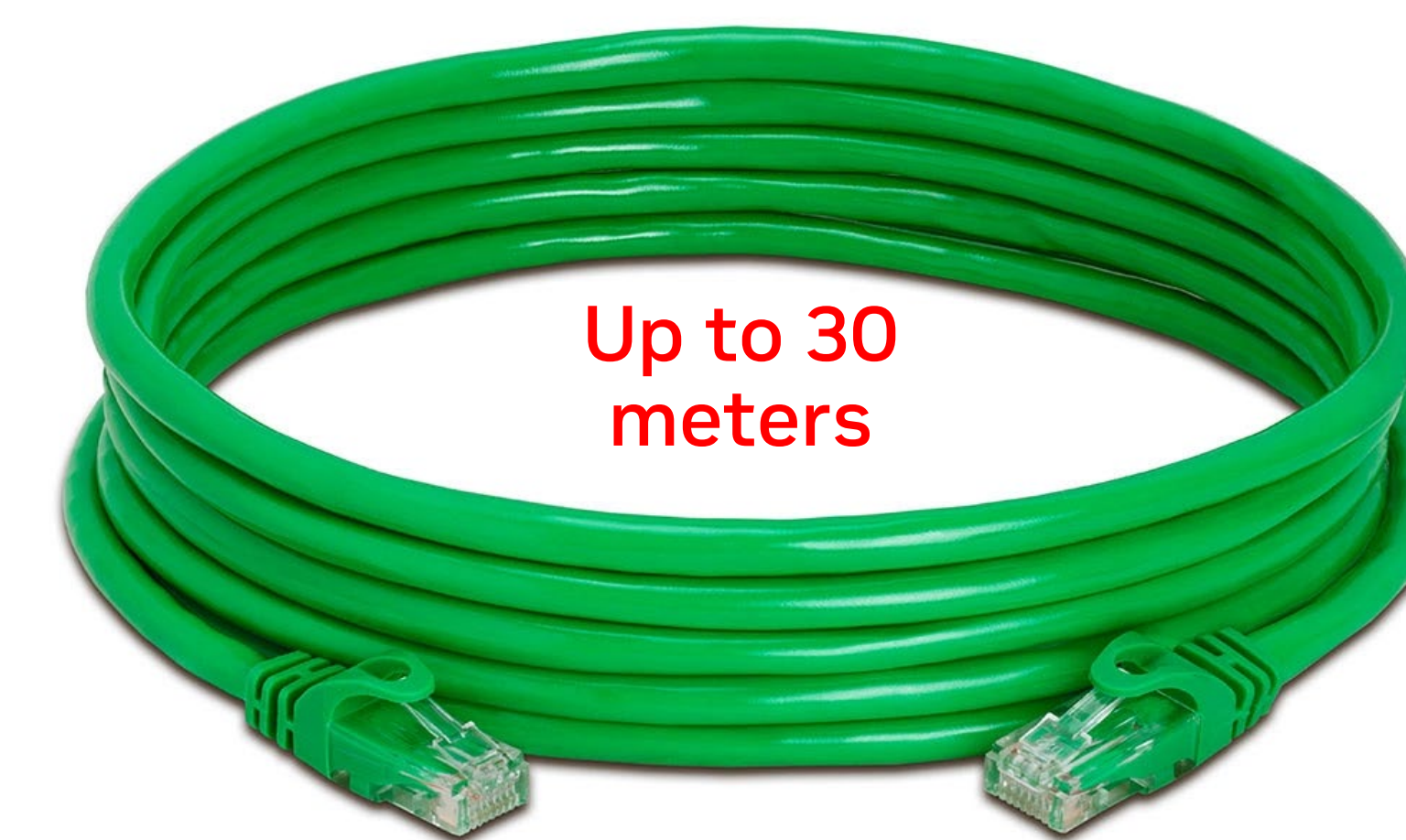
DR4 = Datacenter Reach (500-meters) 4-channels, parallel into MPO-12/APC optical connectors (4 transmit & 4 receive)

FR4= Far Reach (2km) 4-channels, multiplexed into 2-fiber LC optical connectors (1 transmit & 1 receive)



Up to 500
meters

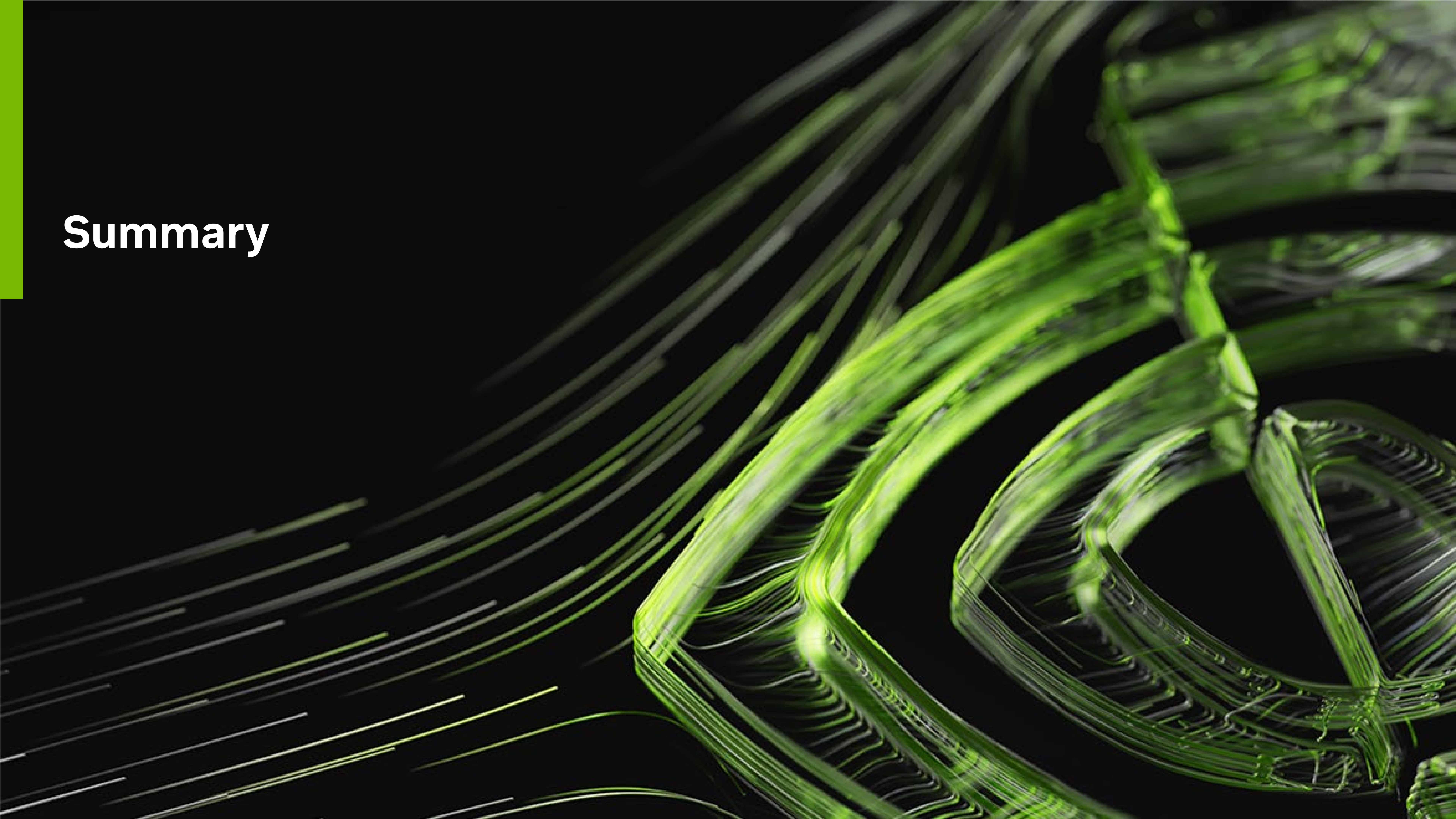
MPO12 APC Single Mode Fiber Optic Cable
0.011 lbs (0.005 kg) per linear foot (typical)
0.12 in (3.0 mm) diameter (typical)



Up to 30
meters

Category 6 Copper Ethernet Cable
0.3 lbs (0.136 kg) per linear foot (typical)
0.295 in (7.55 mm) diameter (SF/UTP) (typical)

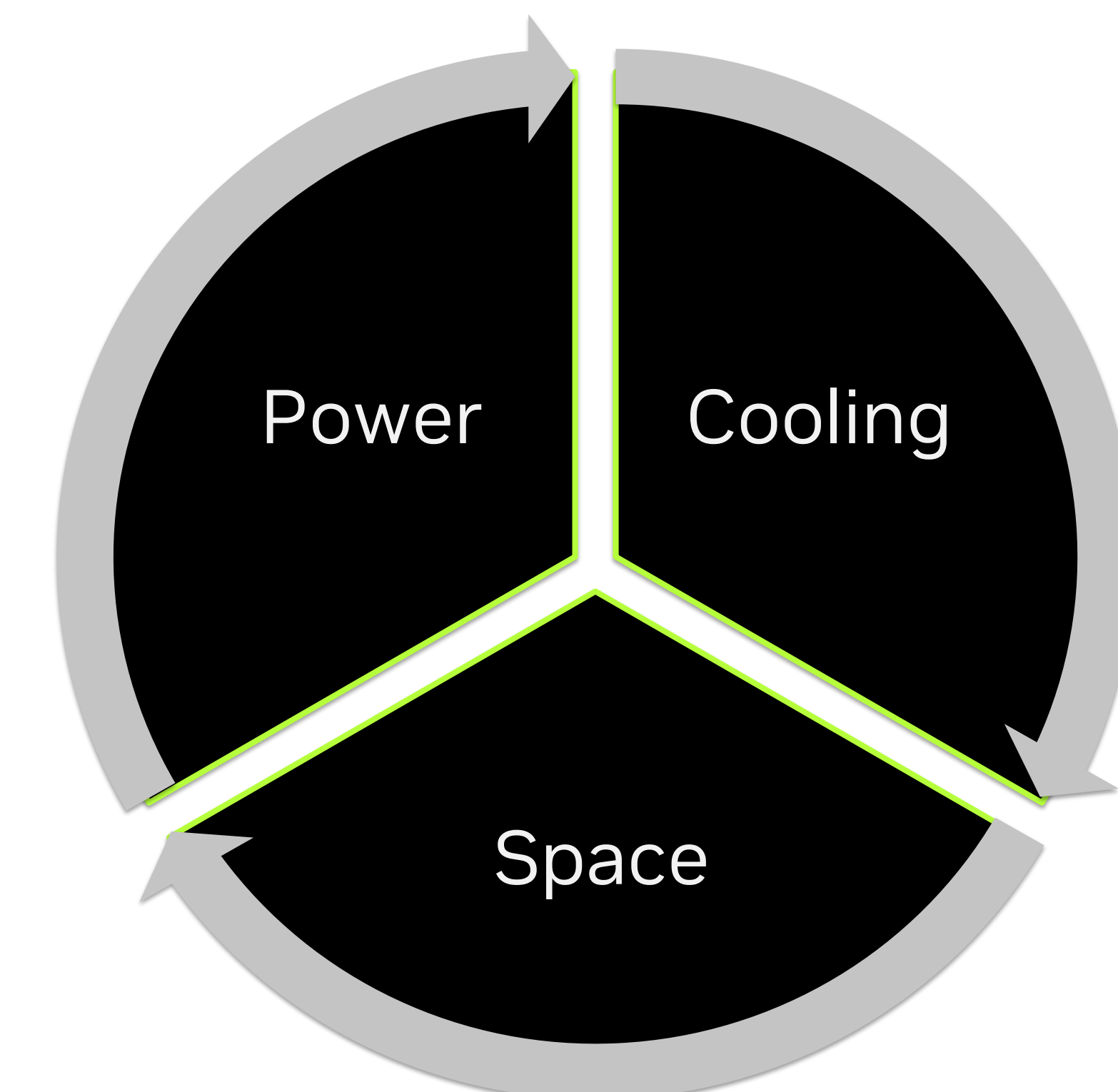
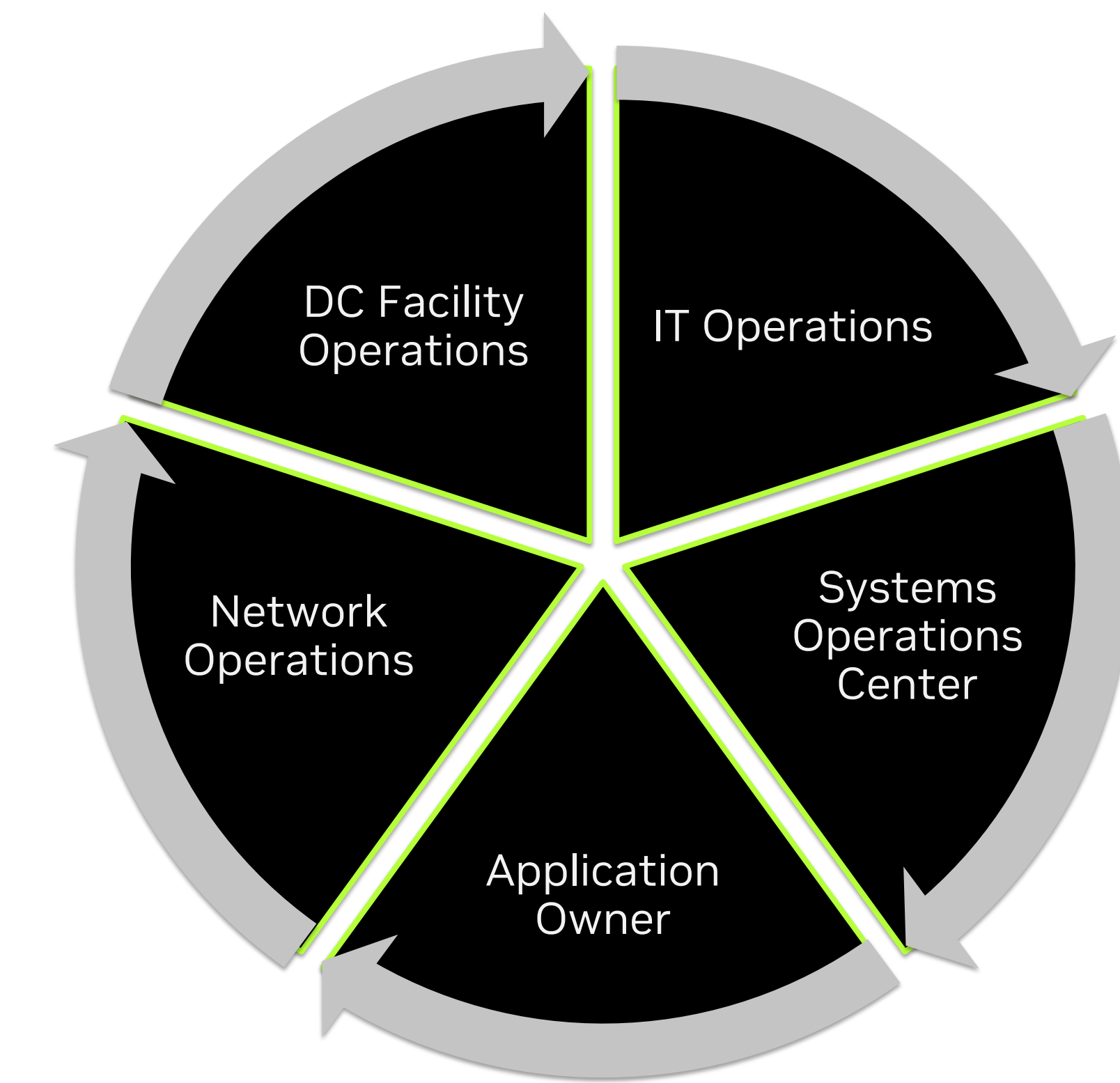
Summary

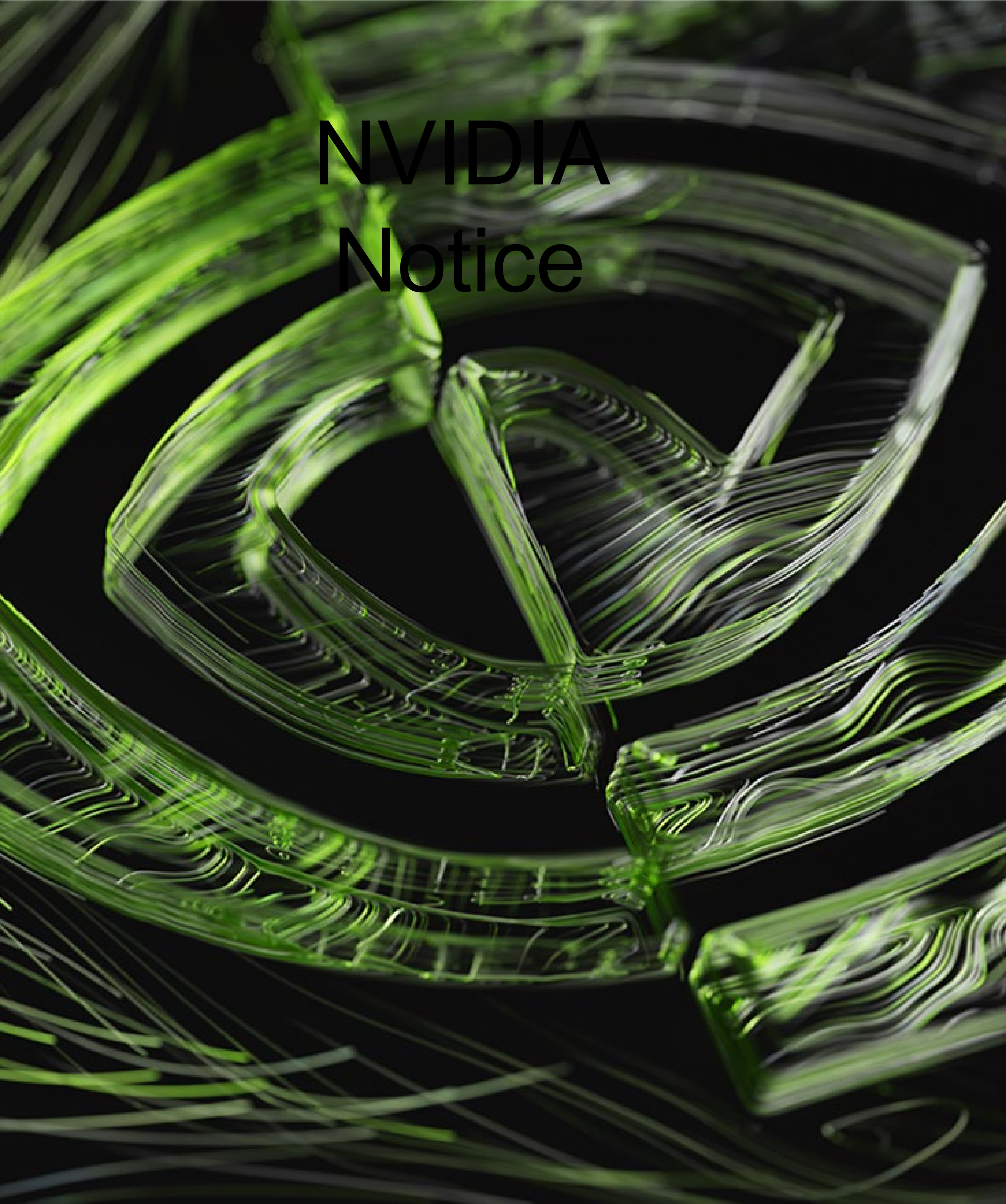


Summary

Align domain owners and plan across a budget of data center resources

- The DGX B300 SuperPOD is a complex system that demands significant power and cooling resources
- Depending on data center site capabilities, this resource demand may require adjacent rows to be unutilized or underutilized, to aggregate power and cooling resources into the SuperPOD rows
- Successful deployments often rely on careful alignment and planning across several organizational and technical domains





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